Supplementary Material for "Evaluating Predictive Utility of Synthetic Liver Cirrhosis Data in IoMT: A Comparative Study of GAN, CTGAN, and TVAE with Consensus-Based Quality Filtering"

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I. Scope of This Document

The main manuscript reports the headline results within a strict page budget, which means a good deal of supporting detail had to be left out. This document supplies that detail. It is organised so that each section can be read on its own, and it is referenced from the main paper as Fig. S1 to Fig. S10 and Table S1 to Table S7.

Three kinds of material appear here. The first is methodological detail that a reader would need in order to reimplement the pipeline exactly: full architecture and optimiser settings, the preprocessing decisions, and the seeding regime that makes the results reproducible. The second is the complete per-feature statistical output. The main paper quotes a handful of Kolmogorov-Smirnov statistics to illustrate a pattern; Tables S2 to S4 give all of them, for every feature and every generative method, so that the pattern can be checked rather than taken on trust. The third is the set of sensitivity and ablation analyses that justify particular parameter choices, chiefly the consensus tolerance of 5.0 and the output-perturbation level of $\sigma = 0.7$.

Nothing in this document changes any conclusion in the main paper. Where a number appears in both, it is the same number, produced by the same seeded run.

II. Extended Methodology

A. Dataset and Preprocessing

The Mayo Clinic Primary Biliary Cirrhosis trial data contains 418 patient records across 20 columns. Patients 1 to 312 were enrolled in the randomised placebo-controlled trial of D-penicillamine; patients 313 to 418 form a later observational cohort who consented to have basic measurements recorded but who did not undergo the full laboratory panel. Each of those 106 records is missing between nine and ten laboratory values.

This matters for how the missingness is handled. Imputing nine or ten values per record would mean inventing the majority of each patient's biochemical profile, and doing so from a cohort measured under a different protocol. The imputed values would carry the statistical structure of the trial arm rather than of the observational arm, which is precisely the kind of systematic bias that would then propagate into the generative models. Complete-case analysis was therefore chosen, leaving 276 records. The cost is a reduction in sample size and a restriction of the analysis to the randomised population; the benefit is that every value used is a value that was actually measured.

The patient identifier column is discarded as it carries no clinical signal. The remaining 19 columns comprise 11 continuous laboratory measurements, five binary clinical indicators, two ordinal variables, and the binary survival outcome. Categorical encodings are given in Section III of the main paper.

All features are scaled to $[0, 1]$ with a MinMaxScaler fitted *only* on the 193-patient training partition. Fitting the scaler on the full dataset before splitting would leak information about the test patients into the training pipeline, and would inflate every downstream result. The same fitted scaler is applied to the test partition and inverted after generation to return synthetic records to clinical units.

B. Generative Architectures

Table S1 gives the complete specification of all three models. All are implemented directly in TensorFlow 2.15 without third-party synthetic-data libraries, which means every architectural choice is visible and auditable rather than inherited from a package default.

The conditional tabular GAN implemented here adopts the outcome-conditioning strategy of Xu et al. but not the full published CTGAN architecture: mode-specific normalisation and training-by-sampling are not implemented. It should therefore be read as a conditional tabular GAN in the spirit of that work, not as a benchmark of CTGAN as originally specified. This is stated in the main paper and repeated here because it is the single most likely source of misreading.

Training curves for all three models are shown in Fig. S2. The adversarial losses settle without the sustained oscillation

TABLE S1
COMPLETE ARCHITECTURE AND OPTIMISER SETTINGS FOR THE THREE GENERATIVE MODELS

Setting	Vanilla GAN	CTGAN	TVAE
Latent dimension	100	100 + 2 (one-hot)	32
Encoder / discriminator	256, 128	256, 128	128, 64
Generator / decoder	128, 256	128, 256	64, 128
Hidden activation	LeakyReLU (0.2)	LeakyReLU (0.2)	ReLU
Output activation	Sigmoid	Sigmoid	Sigmoid
Output dimension	18	18	18
Batch normalisation	Yes (0.8)	Yes (0.8)	No
Dropout	0.3 (disc.)	0.3 (disc.)	No
Optimiser	Adam	Adam	Adam
Learning rate	2×10^{-4}	2×10^{-4}	1×10^{-3}
β_1	0.5	0.5	default
Batch size	32	32	32
Epochs	500	300	300
Conditioning	None	Outcome label	None
Loss	Adversarial	Adversarial	ELBO ($\beta = 1.0$)
Records generated	500	500	500

that signals mode collapse, and the TVAE evidence lower bound falls steeply before levelling off, so all three models reach convergence within their training budgets.

C. Reproducibility

Every result in the main paper and in this document comes from a single seeded execution of `master_runner.py`. Python's `random`, NumPy, and TensorFlow are all seeded at 42, as is the train/test split, every scikit-learn estimator, and the SMOTE resampler. Re-executing the pipeline reproduces all 19 result tables byte for byte, which we verified by checksum after a full clean re-run.

Two practical notes for anyone reproducing this work. First, the consensus procedure used throughout is the *equalised* variant described in Section III of the main paper, in which the TVAE pool is downsampled to match the smallest adversarial pool before voting. An earlier non-equalised variant produces a differently sized consensus set and different downstream numbers. Second, the nearest-neighbour privacy analysis reads the datasets written by `master_runner.py` rather than regenerating them, so it always describes the same synthetic records the rest of the results are computed from.

III. COMPLETE DISTRIBUTIONAL FIDELITY RESULTS

The main paper summarises distributional fidelity with a single aggregate FID per method and quotes a few representative Kolmogorov-Smirnov statistics. That summary hides an important structure, which the full tables make visible: the methods do not fail uniformly across features, and the method with the best aggregate score is not the best on every feature.

A. Kolmogorov-Smirnov Tests

Table S2 gives the two-sample KS statistic between each synthetic pool and the real training data, for all 11 continuous features.

Three observations follow. The TVAE differs significantly from the real data on all 11 features, which is consistent with its poor aggregate FID of 0.2280 and reflects the over-smoothing characteristic of variational reconstruction. The adversarial models each pass on a handful of features and fail badly on the heavily right-skewed ones: both GAN and CT-GAN produce their worst statistics on Bilirubin, Cholesterol and Alkaline Phosphatase, the three most skewed biomarkers in the cohort. This is the expected failure mode, since a sigmoid output layer trained on min-max scaled data has difficulty reproducing a long right tail.

TABLE S2
TWO-SAMPLE KOLMOGOROV-SMIRNOV STATISTICS AGAINST REAL TRAINING DATA. LOWER IS BETTER. AN ASTERISK MARKS A STATISTICALLY SIGNIFICANT DIFFERENCE AT $\alpha = 0.05$

Feature	GAN	CTGAN	TVAE	Consensus
N_Days	0.194*	0.075	0.396*	0.209*
Age	0.083	0.122	0.410*	0.214*
Bilirubin	0.413*	0.487*	0.485*	0.269*
Cholesterol	0.308*	0.454*	0.341*	0.172*
Albumin	0.315*	0.079	0.389*	0.194*
Copper	0.262*	0.286*	0.420*	0.174*
Alk_Phos	0.506*	0.409*	0.423*	0.249*
SGOT	0.165*	0.072	0.425*	0.171*
Triglycerides	0.081	0.179*	0.301*	0.122*
Platelets	0.149*	0.148*	0.405*	0.190*
Prothrombin	0.092	0.116	0.352*	0.133*

The consensus set behaves differently from either of its parents. It is significant on all 11 features, so by the binary reading of the test it looks worse than the individual adversarial pools. But its *magnitudes* are the most uniform of any method, and on the hardest features it is substantially the best: Bilirubin 0.269 against the GAN's 0.413 and CTGAN's 0.487, and Alkaline Phosphatase 0.249 against 0.506 and 0.409. Cross-model agreement trades a small loss on the easy features for a large gain on the difficult ones. This is worth stating carefully, because it is the clearest evidence that consensus voting does what it was designed to do at the distributional level, even though Section IV of the main paper shows it delivers no predictive benefit.

B. Jensen-Shannon Divergence

Table S3 reports the Jensen-Shannon divergence per feature, which is bounded in $[0, 1]$ and, unlike the KS statistic, is sensitive to differences across the whole distribution rather than at the point of maximum separation.

The divergence measure tells a cleaner story than the KS test. The TVAE is between three and ten times worse than every other method on nearly every feature, with N_Days and Age the worst affected at roughly 0.40. The consensus set is the most consistent performer: it never exceeds 0.110 on any feature, whereas the GAN reaches 0.138 on Alkaline Phosphatase and CTGAN reaches 0.150 on Cholesterol. Averaged across features the consensus set is the best of the four, which is the result the aggregate FID of 0.0809 also reflects.

C. Standardised Mean Differences

Table S4 gives Cohen's d between each synthetic pool and the real training data. A positive value means the synthetic mean exceeds the real mean.

TABLE S3
JENSEN-SHANNON DIVERGENCE BETWEEN SYNTHETIC AND REAL TRAINING DISTRIBUTIONS. LOWER IS BETTER

Feature	GAN	CTGAN	TVAE	Consensus
N_Days	0.070	0.039	0.401	0.098
Age	0.056	0.064	0.397	0.110
Bilirubin	0.072	0.077	0.248	0.057
Cholesterol	0.084	0.150	0.140	0.057
Albumin	0.091	0.040	0.304	0.065
Copper	0.063	0.069	0.250	0.071
Alk_Phos	0.138	0.110	0.205	0.080
SGOT	0.033	0.030	0.325	0.062
Triglycerides	0.021	0.048	0.179	0.039
Platelets	0.045	0.055	0.356	0.095
Prothrombin	0.023	0.030	0.219	0.047

TABLE S4
COHEN'S d BETWEEN SYNTHETIC AND REAL TRAINING DATA. VALUES NEAR ZERO INDICATE WELL-MATCHED MEANS

Feature	GAN	CTGAN	TVAE	Consensus
N_Days	-0.245	$+0.100$	$+0.099$	-0.025
Age	-0.012	$+0.007$	-0.072	$+0.076$
Bilirubin	$+0.686$	$+0.695$	-0.016	$+0.530$
Cholesterol	$+0.581$	$+0.661$	-0.044	$+0.426$
Albumin	-0.644	$+0.059$	$+0.163$	-0.193
Copper	$+0.607$	$+0.546$	-0.104	$+0.462$
Alk_Phos	$+0.771$	$+0.667$	-0.025	$+0.531$
SGOT	$+0.362$	$+0.089$	-0.230	$+0.108$
Triglycerides	$+0.229$	$+0.303$	$+0.017$	$+0.281$
Platelets	$+0.248$	$+0.175$	-0.088	$+0.067$
Prothrombin	$+0.159$	$+0.227$	-0.045	$+0.197$

This table contains the most counterintuitive result in the supplementary material, and it deserves attention. By the standardised mean difference criterion the TVAE is the *best* method: every one of its 11 values is negligible, none exceeding 0.230 in magnitude. The adversarial models systematically overshoot on the skewed biomarkers, with the GAN reaching $+0.771$ on Alkaline Phosphatase.

The explanation is that a variational autoencoder trained to minimise reconstruction error will reproduce the mean of the training distribution almost exactly while collapsing its variance and shape. Matching the first moment is not the same as matching the distribution. The KS and Jensen-Shannon results in Tables S2 and S3 show the TVAE failing comprehensively on distributional shape at the same time as it succeeds on means.

The practical lesson generalises well beyond this dataset: a single fidelity metric can be satisfied by a model that is wrong in every other respect, and a method that looks best under one criterion can look worst under another computed on the same data. Reporting one number invites exactly this error. It is the distributional analogue of the main paper's central finding, that distributional similarity and predictive utility are different properties that must be measured separately.

TABLE S5
CONSENSUS TOLERANCE SWEEP. COMPOSITION PERCENTAGES ARE SHARES OF THE ACCEPTED SET. THE REAL TRAINING COHORT IS 40.4% DECEASED

Tol.	Records	GAN %	CTGAN %	TVAE %	Deceased %	FID
2.5	89	11.2	15.7	73.0	1.1	0.2262
3.0	252	20.6	15.1	64.3	10.7	0.1677
3.5	461	21.5	16.1	62.5	17.4	0.1441
4.0	667	22.6	17.4	60.0	21.1	0.1247
4.5	848	23.5	20.8	55.8	23.2	0.1061
5.0	953	25.1	22.6	52.4	24.7	0.0950
5.5	1017	27.4	23.5	49.1	25.8	0.0864
6.0	1043	28.7	23.5	47.8	26.3	0.0829

D. Supporting Figures

Fig. S1 shows the correlation structure of the real cohort, which is the target the generative models are trying to reproduce. Fig. S3 compares marginal distributions for six representative biomarkers, and makes the Bilirubin failure visible directly. Fig. S4 compares the real and consensus correlation matrices, showing that dominant pairwise associations survive generation while weaker ones are attenuated. Fig. S5 projects real and synthetic records into two dimensions and shows the consensus set concentrating in the centre of the real manifold, which is the geometric fact underlying both its good distributional scores and its poor privacy profile.

IV. CONSENSUS THRESHOLD SENSITIVITY

The consensus tolerance of 5.0 standardised units is the single most consequential free parameter in the pipeline, so it was selected by a systematic sweep rather than by inspection. Table S5 reports the sweep across eight values.

The class-balance column is the reason a very tight tolerance is unusable. At 2.5 the accepted set is 1.1% deceased, against 40.4% in the real training cohort. A strict agreement requirement does not sample the outcome classes evenly: deceased patients occupy the sparser, more extreme region of biomarker space, exactly where three independently trained models are least likely to place records near one another. Tightening the tolerance therefore filters out the minority class almost entirely, and augmenting with such a set would worsen the imbalance the augmentation was meant to relieve.

As the tolerance widens, class balance recovers monotonically and FID improves monotonically, but the marginal return falls away. Between 5.0 and 6.0 the accepted set grows by 90 records while FID improves by only 0.0121, and each additional record is by construction one that fewer models agreed on. The value of 5.0 sits at the point where class balance has become tolerable at 24.7% and the quality curve has begun to flatten. It is a defensible compromise rather than an optimum, and we state it as such.

Note that the record counts in this table are from the non-equalised voting procedure, which is why 5.0 yields 953 records here against the 787 reported in the main paper. The equalisation step, which downsamples the TVAE pool before voting, is applied afterwards and reduces the count while improving fidelity.

TABLE S6
GAUSSIAN OUTPUT PERTURBATION APPLIED TO THE FILTERED TVAE POOL ($n = 499$)

σ	Near-dup. (n)	Near-dup. (%)	FID
0.0	178	35.7	0.2280
0.1	169	33.9	0.2095
0.2	140	28.1	0.1720
0.3	88	17.6	0.1402
0.5	29	5.8	0.0804
0.7	7	1.4	0.0481
1.0	2	0.4	0.0319

TABLE S7
ESTIMATED DP-SGD PRIVACY BUDGET AT $\delta = 10^{-5}$ FOR $n = 193$ TRAINING PATIENTS

Noise multiplier	ε	Interpretation
0.5	210.4	Essentially no DP guarantee
0.8	89.2	Very weak privacy guarantee
1.0	61.2	Very weak privacy guarantee
1.2	46.0	Very weak privacy guarantee
1.5	33.6	Very weak privacy guarantee
2.0	23.9	Very weak privacy guarantee
3.0	14.0	Very weak privacy guarantee

V. EXTENDED PRIVACY ANALYSES

A. Output Perturbation

The filtered TVAE pool carries a 35.7% near-duplicate rate, meaning more than one generated record in three sits closer to a real training patient than the 5th-percentile real-to-real distance. Releasing such a set would be indefensible. Table S6 reports a sweep of post-hoc Gaussian perturbation applied to the eleven continuous features, with σ expressed in units of each feature's training standard deviation.

Both columns improve together, which is not what one would expect. Adding noise to synthetic records normally buys privacy at the cost of fidelity; here FID improves from 0.2280 to 0.0481 while the near-duplicate rate falls from 35.7% to 1.4%.

The reason is specific to how the TVAE fails. It does not produce a broadly correct distribution with a few risky records; it produces records clustered tightly around particular real patients, which is simultaneously why its near-duplicate rate is high and why its marginal distributions are too narrow. Perturbation attacks both problems at once, dispersing the clusters away from individual patients and widening the marginals back towards the real spread. The improvement is therefore a correction of a specific defect rather than a general free lunch, and it should not be expected to transfer to a generator whose failure mode is different.

We select $\sigma = 0.7$ rather than 1.0 because it already clears any reasonable near-duplicate target at 1.4%, and pushing further begins to distort a distribution that has by then been corrected.

B. Formal Differential Privacy

Table S7 reports estimated privacy budgets for differentially private stochastic gradient descent at $\delta = 10^{-5}$, across noise multipliers from 0.5 to 3.0, for the $n = 193$ training cohort.

TABLE S8
SUBGROUP TEST-SET AUC, REPORTED AS SCENARIO A / SCENARIO C. SUBGROUPS BELOW $n = 20$ ARE REPORTED FOR COMPLETENESS ONLY

Subgroup	n	Dec.	RF	GB	LR
Early stage (1–2)	17	4	0.558 / 0.577	0.654 / 0.577	0.577 / 0.654
Late stage (3–4)	66	29	0.883 / 0.866	0.924 / 0.902	0.872 / 0.864
Female	72	27	0.840 / 0.835	0.889 / 0.849	0.847 / 0.827
Male	11	6	0.833 / 0.617	0.867 / 0.900	0.667 / 0.833
D-penicillamine	39	17	0.773 / 0.734	0.818 / 0.813	0.824 / 0.861
Placebo	44	16	0.908 / 0.904	0.938 / 0.882	0.855 / 0.848

Values of ε below roughly 10 are usually considered the threshold of a meaningful guarantee, and values below 1 are considered strong. Even at a noise multiplier of 3.0, which would severely damage a generative model trained on 193 records, the budget remains at 14.0. The conclusion is unambiguous and worth stating plainly: formal differential privacy is not attainable on a cohort of this size. The privacy budget scales with the number of training examples, and 193 is simply too few for the accounting to close at a defensible ε.

This is why the main paper recommends output perturbation as the practical mitigation at this scale, and treats DP-SGD as an option that becomes available only once a cohort of several hundred patients has been assembled. Reporting a DP-SGD result at $\varepsilon = 14$ as though it constituted a privacy guarantee would be misleading, and we avoid doing so.

Both analyses in this section are plotted together as Fig. 10 of the main manuscript, which is not reproduced here; the underlying numbers are those in Tables S6 and S7.

VI. SUBGROUP ANALYSIS

Table S8 reports test-set AUC within six clinical strata, comparing the real-only baseline against consensus augmentation. Each cell gives Scenario A followed by Scenario C.

Among the four subgroups large enough to support inference, the picture is consistent with the overall null result. Differences between Scenario A and Scenario C stay within about ± 0.06, with no consistent direction, and every AUC exceeds 0.73. Augmentation neither helps nor harms in any stratum we can measure reliably.

The two small subgroups illustrate why we decline to interpret them. On the 11 male patients, Random Forest AUC moves from 0.833 to 0.617 while Gradient Boosting moves from 0.867 to 0.900 on the same patients under the same augmentation. Two classifiers cannot both be right about the direction of an effect; with six deceased patients in the subgroup, a single reordered prediction shifts AUC by several points. The early-stage subgroup, with 17 patients and four deceased, is equally unstable and additionally sits near chance for two of the three classifiers.

We report these rows rather than dropping them because selective omission of inconvenient strata is itself a reporting problem. But no weight should be placed on them, and a reader who wishes to cite a subgroup result should use the late-stage, female, or treatment-arm rows.

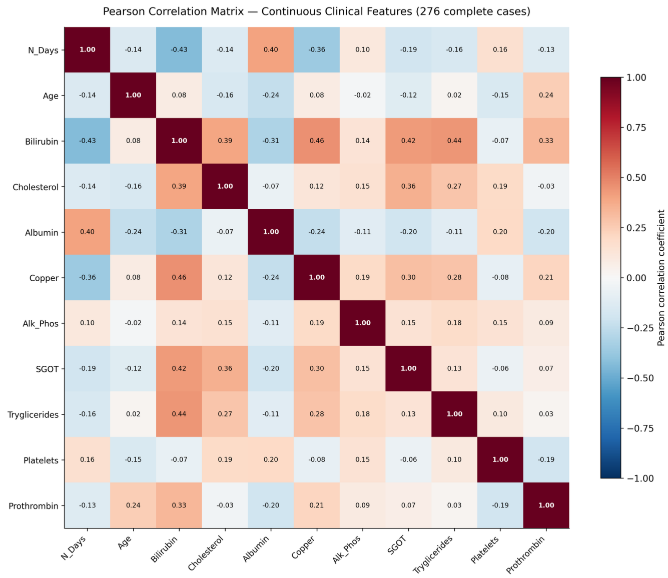

Fig. S1. Pearson correlation matrix across all 276 complete cases. Bilirubin correlates positively with Copper ($r = 0.46$), SGOT ($r = 0.42$), Cholesterol ($r = 0.39$) and Prothrombin ($r = 0.33$), reflecting the interlinked nature of hepatic synthetic and excretory function in PBC. N_Days correlates negatively with Bilirubin ($r = -0.43$) and Copper ($r = -0.36$), consistent with more severe biochemical profiles being associated with shorter survival.

VII. Graphical Views of the Predictive Results

Five figures present the predictive results in graphical form: Precision-Recall curves (Fig. S6), AUC across scenarios (Fig. S7), the F1 heatmap (Fig. S8), and grouped metric comparisons (Figs. S9 and S10). Every quantity plotted appears numerically in Tables III and IV of the main manuscript, so these figures add no new evidence; they are provided because the pattern across scenarios is easier to see at a glance than to reconstruct from a table.

The pattern is the same in all five. Every scenario and classifier combination clears AUC = 0.8, so no augmentation strategy damages discrimination. No augmentation scenario uniformly improves on the baseline either. Gradient Boosting is the only classifier whose F1 declines monotonically as synthetic volume increases from Scenario A through Scenario C, which is consistent with the argument in Section IV-H of the main paper that a boosting model fitted on a sufficient real signal has least to gain and most to lose from additional records drawn from the centre of the distribution.

VIII. Supplementary Figures

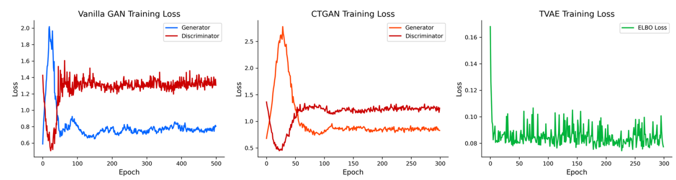

Fig. S2. Training loss curves. Left: Vanilla GAN generator and discriminator losses over 500 epochs. Centre: conditional tabular GAN losses over 300 epochs. Right: TVAE evidence lower bound over 300 epochs. The adversarial losses settle without sustained oscillation, indicating that mode collapse did not occur, and the TVAE objective levels off well before its epoch budget is exhausted.

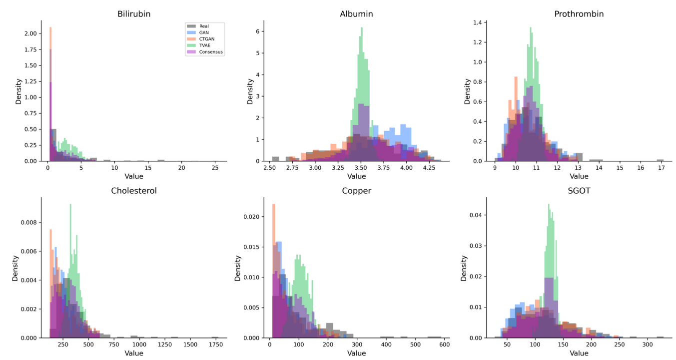

Fig. S3. Marginal distributions of six representative biomarkers for real training data and all four synthetic datasets. GAN and CTGAN reproduce Albumin and Prothrombin closely. All methods diverge on the heavily right-skewed Bilirubin and Cholesterol, and the TVAE distribution for Bilirubin is shifted noticeably rightward, matching the KS statistics in Table S2.

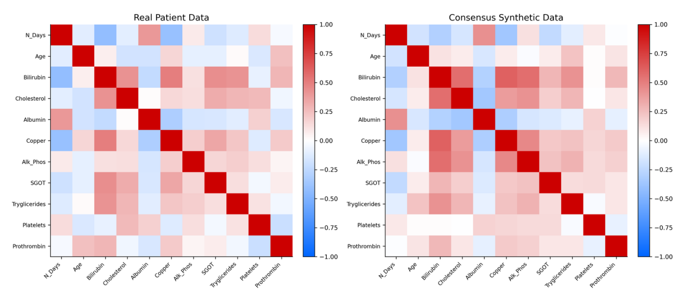

Fig. S4. Pearson correlation structure of real training data (left) against the equalised consensus set (right). Dominant clinical associations are preserved, including the positive Bilirubin-Copper and negative N_Days-Bilirubin relationships, while weaker correlations are attenuated.

Fig. S5. t-SNE projection of real patients and each synthetic method from the original 18-dimensional feature space. Filtered GAN and CTGAN records overlap the real manifold. TVAE records spread beyond the real density contours. The consensus set concentrates in the central high-density region, which explains both its favourable distributional scores and its elevated near-duplicate rate of 21.1% reported in Table II of the main paper.

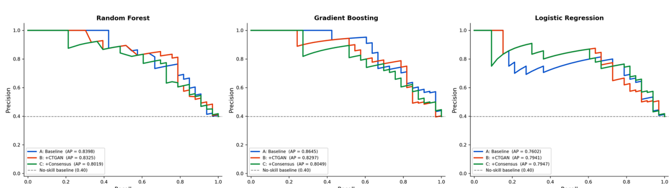

Fig. S6. Precision-Recall curves by classifier and training scenario on the 83-patient real held-out test set. PR curves are informative for this cohort given its class imbalance of 78 deceased against 115 survivors in the training set. All classifiers hold precision above 0.7 across a wide recall range.

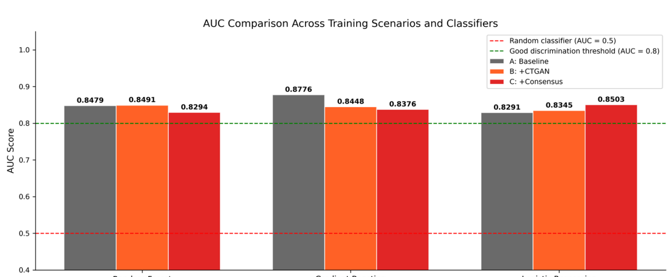

Fig. S7. AUC across all three classifiers and training scenarios A through D. The green dashed line marks AUC $= 0.8$ and the red dashed line marks AUC $= 0.5$. Every combination clears 0.8. Gradient Boosting under Scenario A achieves the highest AUC at 0.878.

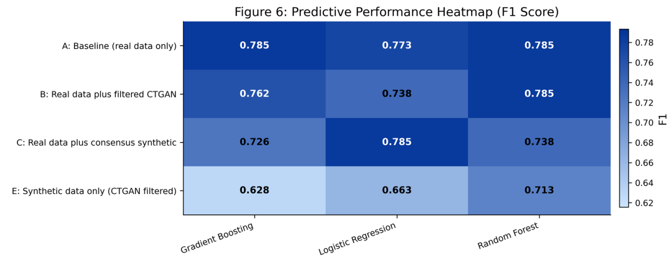

Fig. S8. F1-score heatmap across training scenarios; darker shading indicates higher F1. Gradient Boosting under Scenario D achieves the highest F1 at 0.809. Under Scenario C, Gradient Boosting falls to 0.726 and Random Forest to 0.738 while Logistic Regression rises to 0.785, showing that consensus augmentation does not move all classifiers in the same direction.

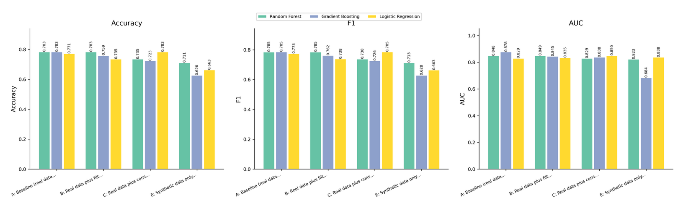

Fig. S9. Grouped comparison of accuracy, F1 and AUC across all training scenarios and classifiers. No augmentation scenario uniformly outperforms the real-data baseline.

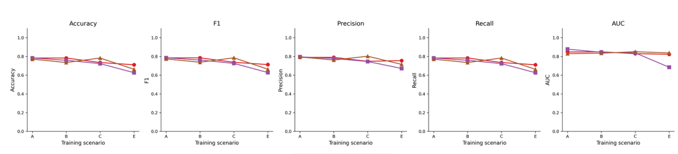

Fig. S10. All five evaluation metrics across training scenarios for each classifier. Trajectories are flat or slightly declining from Scenario A through Scenarios B and C, with Scenario D producing a modest accuracy gain for Gradient Boosting only.

IX. Data and Code Availability

The Mayo Clinic PBC dataset is publicly available from the UCI Machine Learning Repository. All code required to reproduce every number and figure in the main paper and in this document is available at https://github.com/Michaeludousoro/liver-cirrhosis-synthetic-data, including the generative models, the IQR and consensus filters, the evaluation pipeline, and the scripts that produce each supplementary table.